

RPS ARENA

Rock Paper Scissors Game

Project Type: Java GUI Application

Programming Language: Java

IDE: Apache NetBeans

Interface: Graphical User Interface (GUI)

1. Introduction

RPS Arena is a simple Rock Paper Scissors game developed using Java. The project provides a graphical interface where the player can choose Rock, Paper, or Scissors. The computer makes a random choice, and the game determines the winner.

2. Project Objectives

To develop a simple game using Java.

To create a user-friendly GUI.

To implement Rock Paper Scissors game logic.

To generate random choices for the computer.

To maintain and display game scores.

To allow the player to play multiple rounds.

3. Project Features

Main menu

Play Game option

Rock, Paper and Scissors buttons

Random computer choice

Winner and draw detection

Player and computer score tracking

Multiple-round gameplay

Play Again option

Game instructions

Exit option

4. Tools and Technologies

Programming Language: Java

IDE: Apache NetBeans

GUI: Java Swing

Development Platform: Java JDK

5. Game Structure

The game follows this basic process:

Start



Main Menu



Play Game



Player Selects Rock/Paper/Scissors



Computer Makes Random Choice



Compare Choices



Display Result



Update Score



Play Again

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Exit

6. Implementation

The project was implemented using Java Swing for the graphical user interface. Buttons are used for player input. The Random class is used to generate the computer's choice. Conditional statements are used to determine the winner.

The game also keeps track of:

Player Wins

Computer Wins

Draws

7. Testing

The project was tested by:

Testing all three player choices.

Testing win, lose and draw situations.

Testing multiple rounds.

Testing score calculations.

Testing the Play Again option.

Testing the Main Menu and Instructions.

Checking the game for errors.

8. Project Outcome

The completed RPS Arena provides a functional graphical Rock Paper Scissors game. The player can play multiple rounds against the computer and see the updated score after each round.

9. Conclusion

RPS Arena successfully demonstrates the use of Java programming, GUI development, random selection, conditional logic, score management and game testing. The project provides a simple and interactive gaming experience. Project Status: Completed